# Sine-Generated Shifted Lindley Distribution: Properties and Applications

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2. SGSL Distribution and Validity
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4. MGF,Quantile and Characteristic Functions
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# Introduction

- Proposes Sine-Generated Shifted Lindley (SGSL) distribution.
- Applies sine transformation to shifted Lindley CDF for flexibility.
- Enhances skewness and kurtosis for complex data modeling.
- Applications: Reliability, actuarial science, biomedical, environmental.

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# SGSL Distribution

- Shifted Lindley PDF and CDF:

$$f(x;\theta,\mu) = \frac{\theta^2}{1+\theta(1+\mu)} \exp\{-\theta(x-\mu)\}, \quad x > \mu > 0$$

$$F(x;\theta,\mu) = 1 - \frac{1+\theta(1+x)}{1+\theta(1+\mu)} \exp\{-\theta(x-\mu)\}$$

- SGSL CDF and PDF:

$$G_{\text{SGSL}}(x;\theta,\mu) = \sin\left(\frac{\pi}{2}F(x;\theta,\mu)\right)$$

$$g_{\text{SGSL}}(x;\theta,\mu) = \frac{\pi}{2}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right)\frac{\theta^2}{1+\theta(1+\mu)}\exp\{-\theta(x-\mu)\}$$

# Validity of SGSL

- Non-decreasing:

$$\frac{d}{dx} G_{\text{SGSL}}(x;\theta,\mu) = \frac{\pi}{2}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right) f(x;\theta,\mu) \geq 0$$

- Limits:

$$\lim_{x\to 0} G_{\text{SGSL}}(x;\theta,\mu) = 0, \quad \lim_{x\to\infty} G_{\text{SGSL}}(x;\theta,\mu) = 1$$

- Right-continuous: Ensured by continuity of sine and shifted Lindley CDF.

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# Moments and Properties

- Mean: $E[X] \approx \frac{\pi}{2}\mu_X - \frac{\pi^3}{8}\int_\mu^\infty x\left(1 - \frac{1+\theta(1+x)}{1+\theta(1+\mu)}\exp(-\theta(x-\mu))\right)^2 f(x;\theta,\mu)\,dx$
- $E[X^2] \approx \frac{\pi}{2}\mu_{X^2} - \frac{\pi^3}{8}\int_\mu^\infty x^2\left(1 - \frac{1+\theta(1+x)}{1+\theta(1+\mu)}\exp(-\theta(x-\mu))\right)^2 f(x;\theta,\mu)\,dx$
- Variance: $\mathrm{Var}[X] = E[X^2] - (E[X])^2$
- Skewness: $\gamma_1 \approx \frac{1}{\sigma^3}\int_\mu^\infty (x-\mu)^3 \frac{\pi}{2}\left[1 - \frac{1}{2}\left(\frac{\pi}{2}F(x;\theta,\mu)\right)^2\right] f(x;\theta,\mu)\,dx$
- Kurtosis: $\gamma_2 \approx \frac{1}{\sigma^4}\int_\mu^\infty (x-\mu)^4 \frac{\pi}{2}\left[1 - \frac{1}{2}\left(\frac{\pi}{2}F(x;\theta,\mu)\right)^2\right] f(x;\theta,\mu)\,dx$

# Shape Analysis

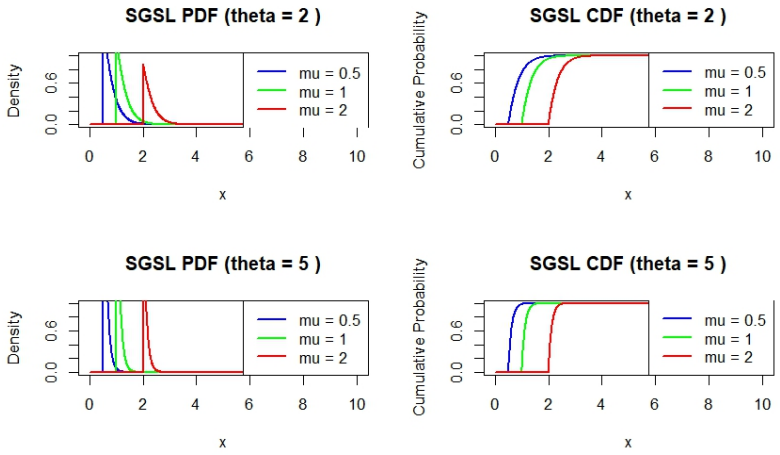


Figure: SGSL PDF and CDF for $\theta = 2, 5,\ \mu = 0.5, 1, 2$.

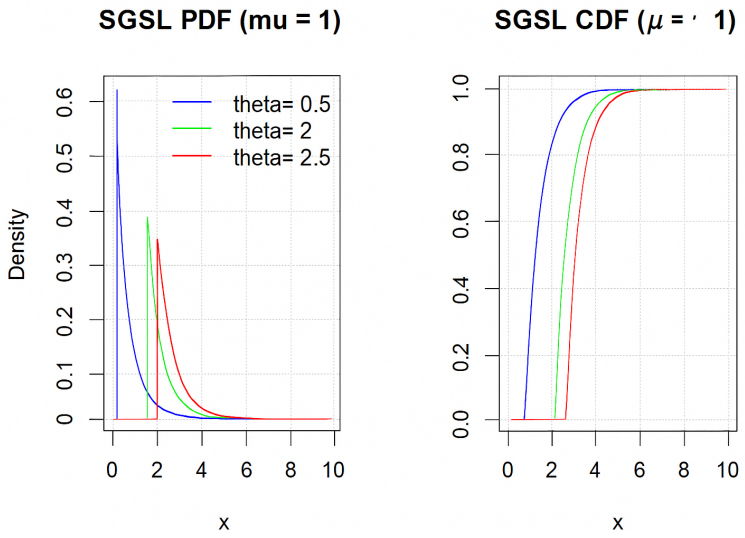


Figure: SGSL CDF and PDF for $\theta = 0.5, 2, 2.5,\ \mu = 1$.

- Varying $\mu$:
  Shifts distribution rightward.
  For $\theta = 5$, the distribution becomes bimodal for smaller $\mu$ and then transitions to unimodal as  increases, with the peak shifting rightward and the overall variance de- creasing.
- Varying $\theta$ :
  Higher $\theta$ (e.g.,2.5): Sharper peaks, reduce dispersion compare to small $\theta$.
  Lower $\theta$ (e.g., 0.5): Heavier tails, more variability(Possible of observations more spread out).

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# MGF,Quantile and Characteristic Functions

- Moment Generating Function:

$$M_{\mathrm{SGSL}}(t;\theta,\mu) \approx \frac{\pi\theta^2 \exp(\theta\mu)}{2(1+\theta(1+\mu))}\left(\frac{1}{\theta-t} - \frac{\pi^2}{8(3\theta-t)}\right)$$

- Quantile Function (numerical):

$$\ln\left(\frac{1+\theta(1+x)}{1+\theta(1+\mu)}\right) = \ln\left(1-\frac{2}{\pi}\arcsin(p)\right) + \theta(x-\mu)$$

- Characteristic Function:

$$\varphi_{\mathrm{SGSL}}(t;\theta,\mu) = \frac{\pi\theta^2 \exp(\theta\mu)}{2[1+\theta(1+\mu)]}\int_{\mu}^{\infty} e^{itx}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right)\exp(-\theta$$

(x - $\mu$))$dx$

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# Reliability Characteristics

- Reliability Function:

$$R_{\text{SGSL}}(x;\theta,\mu) = 1 - \sin\left(\frac{\pi}{2}F(x;\theta,\mu)\right)$$

- Hazard Function:

$$h_{\text{SGSL}}(x;\theta,\mu) = \frac{\frac{\pi}{2}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right)\frac{\theta^2}{1+\theta(1+\mu)}\exp(-\theta(x-\mu))}{1-\sin\left(\frac{\pi}{2}F(x;\theta,\mu)\right)}$$

- Mean Time to Failure:

$$\text{MTTF} = \frac{\pi\theta^2}{2(1+\theta(1+\mu))}\int_{\mu}^{\infty} x\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right)\exp(-\theta(x-\mu))$$

dx

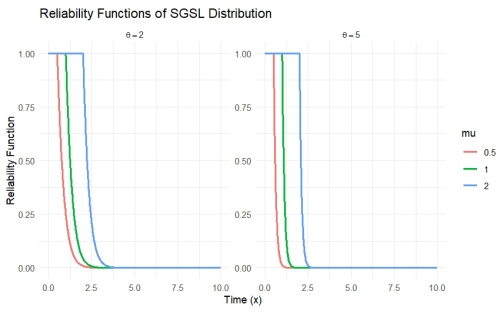


Figure: Reliability function for SGSL $\theta = 2, 5$

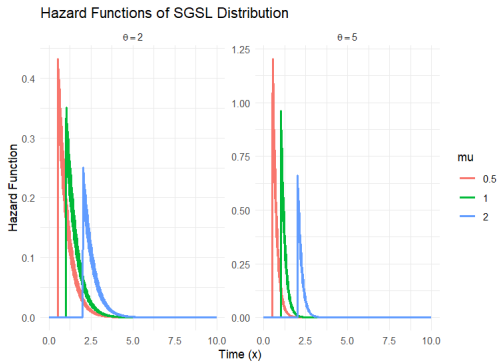


Figure: Hazard rate for SGSL $\theta = 2, 5$,

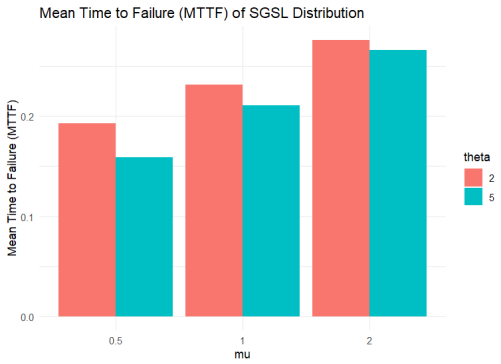


Figure: MTTF of SGSL $\theta = 2, 5$

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# Parameter Estimation

- Log-likelihood:

$$\ell(\theta,\mu) = n\log\left(\frac{\pi}{2}\right) + \sum_{i=1}^{n}\log\cos\left(\frac{\pi}{2}F(x_i;\theta,\mu)\right) + 2n\log(\theta)$$

- n $\log(1+\theta(1+\mu)) - \theta\sum_{i=1}^{n}(x_i-\mu)$

- $\theta$ unknown, solve:

$$\frac{\partial\ell}{\partial\theta} = -\sum_{i=1}^{n}\frac{\pi}{2}\tan\left(\frac{\pi}{2}F(x_i;\theta,\mu)\right)\frac{\partial F}{\partial\theta} + \frac{2n}{\theta} - \frac{n(1+\mu)}{1+\theta(1+\mu)}$$

- $\sum_{i=1}^{n}(x_i-\mu) = 0$

- The function is not theoretically solvable because it is non-linear, so we use Newton-Raphson's algorithm.
- The method stops when $|\theta_{k+1} - \theta_k|$ is very small.
- $\mu$ unknown:

$$\hat{\mu} = \min(x_1, x_2, \ldots, x_n)$$

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# Simulation

- Inverse Transform Sampling:

$$x = G_{\text{SGSL}}^{-1}(u;\theta,\mu) = F^{-1}\left(\frac{2}{\pi}\arcsin(u);\theta,\mu\right), \quad u \sim \text{Uniform}[0,1]$$

- Validation: 1000 samples match theoretical PDF/CDF for $\theta = 2.5$, $\mu = 0.75$.

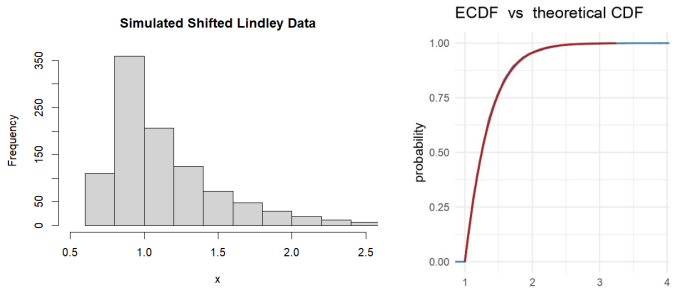


Figure: Histogram and ECDF of SGSL samples.

- **Initial Parameters:**
- $\theta = 2.5$
- $\mu = 0.5$
- **Estimated Parameters:**
- $\hat{\theta} = 2.7827$
- $\hat{\mu} = 0.7502107$
- **Bias:**
- $\text{Bias}(\hat{\theta}) = \hat{\theta} - \theta = 2.7827 - 2.5 = 0.2827$
- $\text{Bias}(\hat{\mu}) = \hat{\mu} - \mu = 0.7502107 - 0.75 = 0.0002107$

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# Data Analysis

- Dataset:The dataset consists of the estimated time from growth hormone medication until the children reached the target age (Alizadeh et al., 2018).
- Statistics: Mean = 5.31, Variance = 8.48, Skewness = 1.37, Kurtosis = 1.40.

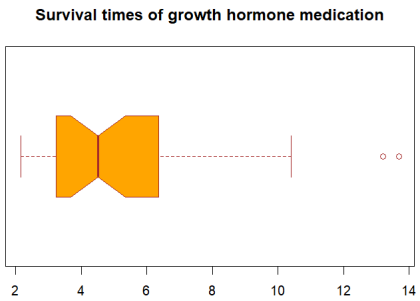


Figure: Box Plot

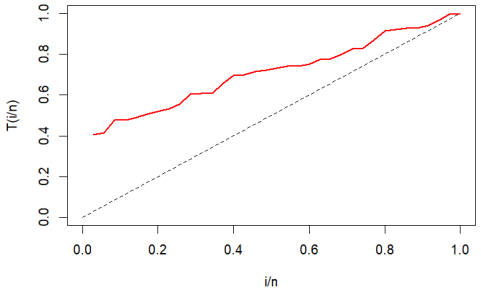


Figure: TTT Plot

Model Comparison:

| Model | Parameters | AIC | BIC | Log-likelihood |
|---|---|---|---|---|
| SGSL | $\hat{\theta} = 0.4102, \hat{\mu} = 2.15$ | 162.34 | 161.42 | $-79.17$ |
| Shifted-Lindley | $\hat{\theta} = 0.434, \hat{\mu} = 2.15$ | 277.00 | 269.91 | $-136.50$ |
| Lindley | $\hat{\theta} = 0.3225$ | 186.68 | 188.24 | $-92.34$ |
| Exponential | $\hat{\theta} = 0.1885$ | 188.81 | 190.37 | $-93.41$ |

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# Conclusions

- SGSL extends shifted Lindley with enhanced flexibility.
- Low AIC and BIC(AIC=162.34, BIC=161.42) tells SGSL model fits the data .
- Suitable for increasing hazard rates.
- Future work: Refine numerical methods, explore new applications.

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# Applications

- **Reliability Engineering**: Models time to failure of systems.
- **Actuarial Science**: Models claim sizes, inter-claim times.
- **Environmental Studies**: Models rainfall, pollutant levels.
- **Biomedical Sciences**: Models survival times, event occurrences.

**THANK YOU**